

Public Finance
BIEM – Spring 2026
Taxation (Ch. 18)



Taxation in this course

18.1 Types of Taxation

18.3 Measuring the Fairness of Tax Systems

18.4 Defining the Income Tax Base

18.5 Externality/Public Goods Rationales for Deviating from Haig-Simons

18.6 The Appropriate Unit of Taxation



Basic concepts of taxation

- Tax base (or taxable income): Y
- Total amount of taxes: T
- Average tax rate (ATR): the percentage of total income that is paid in taxes
- Marginal tax rate (MTR): the percentage that is paid in taxes of the next euro earned

$$\bar{t} = \frac{T}{Y} \text{ average tax rate} \quad t' = \frac{\partial T}{\partial Y} \text{ marginal tax rate}$$

- Statutory tax rates: tax rates laid out in the legal schedule
- Effective tax rate: tax rates an individual actually pays (due to exemptions and deductions the taxable income base may be lower than total income)



Tax progressivity: definitions

- A tax is proportional if the average tax rate does not change with the tax base: all taxpayers pay the same proportion of their income in taxes
- A tax is progressive if the average tax rate increases when the tax base increases: higher marginal tax rates are applied to higher tax bases ($MTR > ATR$)
- A tax is regressive if the average tax rate decreases when the tax base increases



Tax progressivity

Tax on income T is a function of the income Y

$$\bar{t} = \frac{T}{Y} \text{ average tax rate} \quad t' = \frac{\partial T}{\partial Y} \text{ marginal tax rate}$$

How the average tax rate changes with the income

$$\frac{\partial \bar{t}}{\partial Y} = \frac{\partial(\frac{T}{Y})}{\partial Y} = \frac{t'Y - T}{Y^2} = \frac{t' - \bar{t}}{Y}$$

The tax is progressive if $\frac{\partial \bar{t}}{\partial Y} > 0$

Which is equivalent to say that $t' > \bar{t}$

Progressivity by tax credits

$$T = tY - f$$

$$t' = t$$

Taxpayers are allowed to reduce the taxes by a certain amount f (spending, for example, on child care).

$$\bar{t} = \frac{T}{Y} = \frac{tY - f}{Y} = t - \frac{f}{Y} \Rightarrow \bar{t} < t$$

Progressivity by deductions

$$T = t(Y - d)$$

$$t' = t$$

A taxpayer can deduct a fixed amount d from his taxable income (ex. spending on charitable donations or home mortgage interest)

$$\bar{t} = \frac{T}{Y} = \frac{t(Y - d)}{Y} = t - \frac{td}{Y} \Rightarrow \bar{t} < t$$

Which is more redistributive?



Comments on progressivity

- Tax brackets, deductions and tax credits are typically all present, but one instrument is enough to reach progressivity
- There is a trend in reduction of the number of tax brackets:
 - Average number in OECD countries: 14 (1980), 6 (1990), 5 (2000)
- The top marginal tax rate is an important measure



Taxation across income levels

Tax systems typically apply higher average tax rates at higher income levels

Progressivity reflects:

- graduated income tax schedules
- social security contribution caps
- tax credits and allowances at low incomes

As a result:

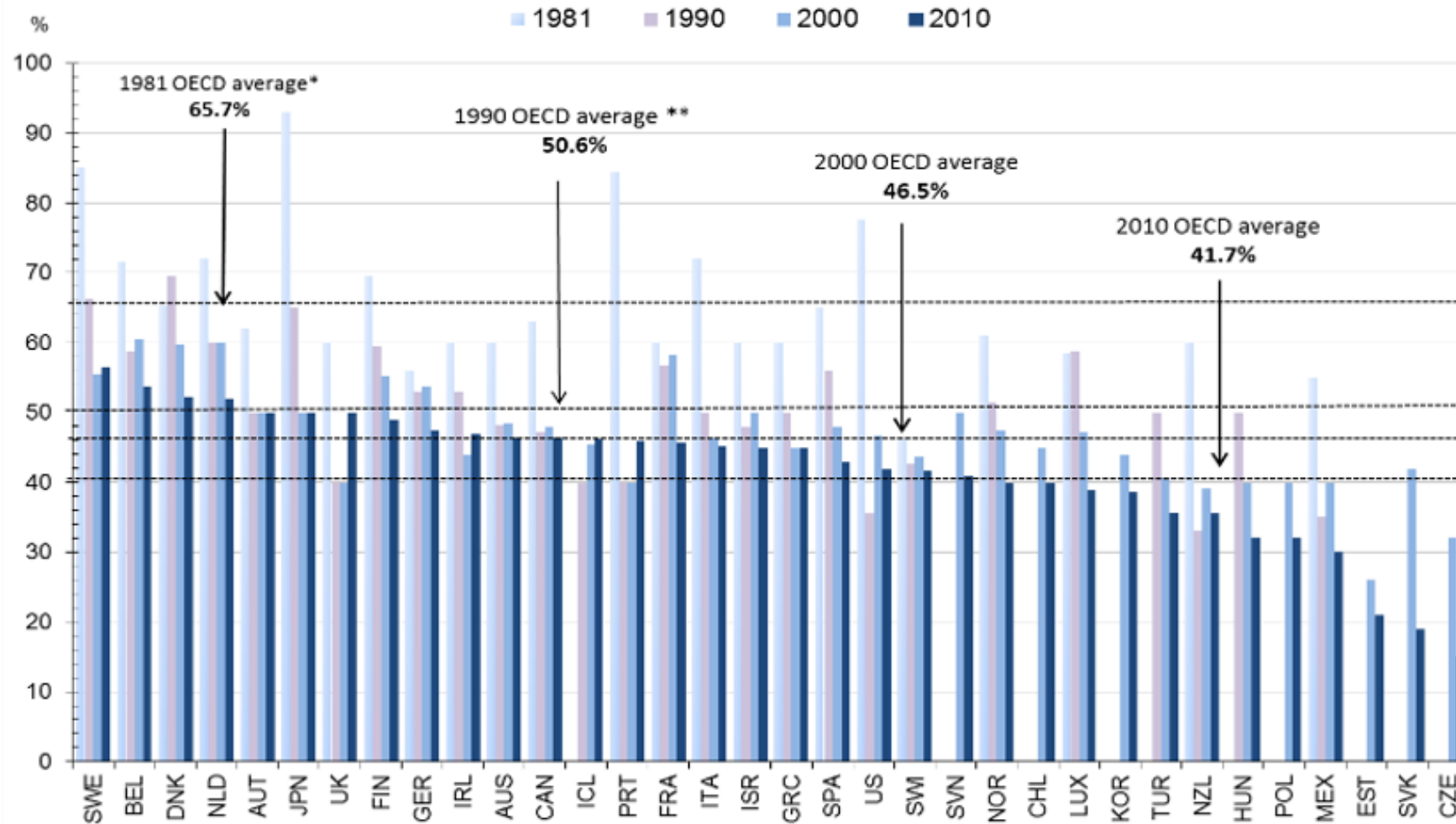
- redistribution is mainly driven by average tax rates
- behavioral responses depend on marginal tax rates

The degree of progressivity varies substantially across OECD countries

Source: OECD (2024), *Taxing Wages*

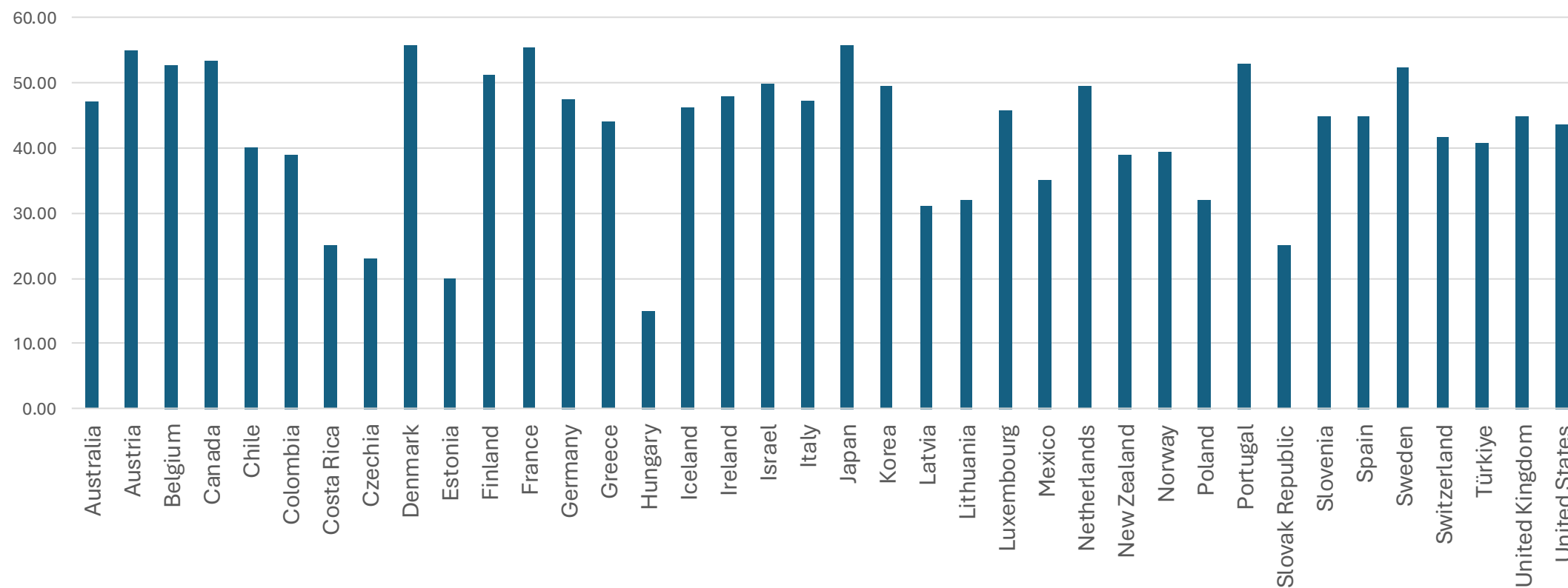


Maximum tax rate, personal income tax, OECD countries



Maximum tax rate, personal income tax, OECD countries

2022



Tax progressivity

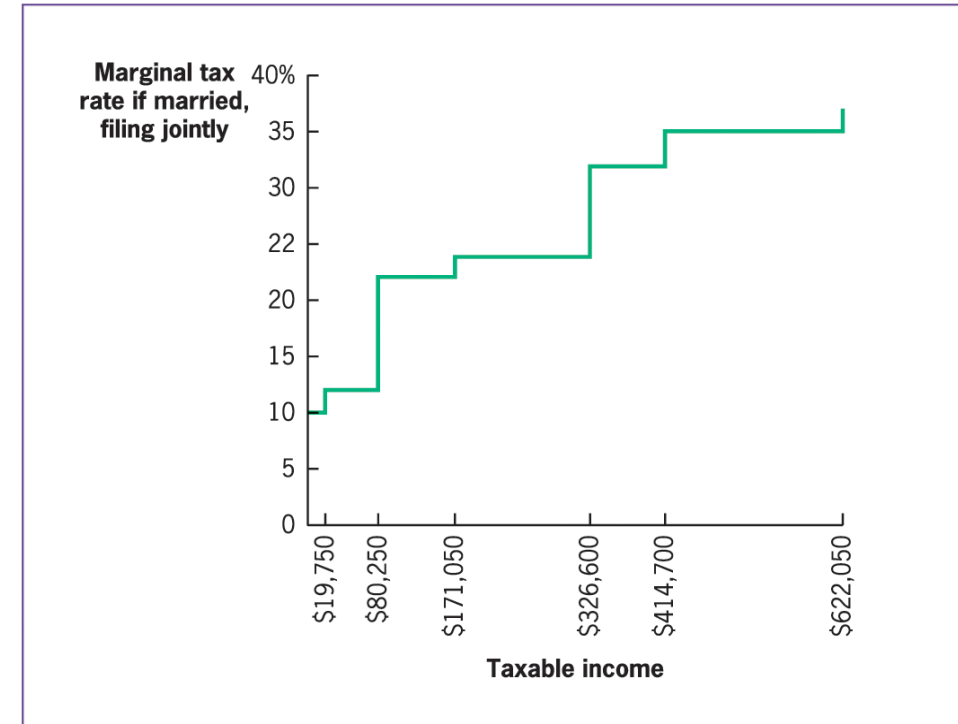
- Direct taxes are typically progressive
 - However: taxes on labour income are progressive, while taxes on capital income are typically not progressive and they are low
- Indirect taxes (consumption) may be even regressive
- Contributions are typically proportional (although floors and ceilings on taxable earnings often apply)

Measuring the Fairness of Tax Systems 1

Two key features of any tax system:

- **Marginal tax rate:** The percentage of the next dollar of taxable income that is paid in taxes.
 - In the US, the marginal tax rate rises with income, from 10% to 37%.
- **Average tax rate:** The percentage of gross income that is paid in taxes:

$$\frac{\text{Total Tax Payments}}{\text{Total Income}}$$



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Measuring the Fairness of Tax Systems 2

Example: Ranya has

Gross income =\$200,000

Taxable income =\$180,000

Total Tax Bill =

$(19,750 \times 0.1) + (60,500 \times 0.12) + (90,800 \times 0.22) + (8,950 \times 0.24) = \$31,359$

Her **marginal tax rate** is 24%

Her **average tax rate** is

$$\frac{\text{Total Tax Payments}}{\text{Total Income}} = \frac{31,359}{200,000} = 15.7\%$$



Vertical and Horizontal Equity 1

Two distributional goals are considered in measuring tax fairness:

- **Vertical equity:** The principle that groups with more resources should pay higher taxes than groups with fewer resources.
- **Horizontal equity:** The principle that similar individuals who make different economic choices should be treated similarly by the tax system.



Vertical and Horizontal Equity 2

Extreme example for the horizontal equity:

- State 1 sets a sales tax of 5% on all goods.
- State 2 implements a system whereby whenever you make a purchase, the cashier flips a coin, and you pay no sales tax if it is heads and a 10% sales tax if it is tails.

State 2's system will raise the same amount of money as State 1's system on average. However, it is much less horizontally equitable because two identical individuals could end up paying very different taxes.



Measuring Vertical Equity

Vertical equity likely requires progressive taxation.

- **Progressive:** Tax systems in which effective average tax rates rise with income.
 - *Example:* individuals pay 10% of their income in tax at an income of \$10,000, but they pay 30% of their income in tax at an income of \$100,000.
- **Proportional:** Tax systems in which effective average tax rates do not change with income so that all taxpayers pay the same proportion of their income in taxes.
 - *Example:* individuals pay 15% of their income in taxes regardless of whether they earn \$10,000 or \$100,000.
- **Regressive:** Tax systems in which effective average tax rates fall with income.
 - *Example:* individuals pay 15% of their income in tax at an income of \$10,000, but pay only 10% of their income in tax at an income of \$100,000.



Defining the Income Tax Base: The Haig–Simons Comprehensive Income Definition

What kind of income should be taxed?

- Haig–Simons comprehensive income definition: Defines taxable resources as the change in an individual's power to consume during the year.
- Potential annual consumption: Total consumption during the year, plus any increases in wealth.
- Difficulties:
 - Defining power to consume/ability to pay
 - Nonconsumption expenditures



Defining the Income Tax Base: Deviations

Haig–Simons promotes horizontal equity by treating all income equally.

- One reason for deviation is to account for expenses not associated with desired consumption.
 - Deductions for property and casualty losses
 - Deduction for medical expenditures
 - Deduction for state and local tax payments
- Another reason for deviation is that some expenditures are not for consumption but rather reflect the cost of earning a living.
 - Legitimate costs of doing business should be deducted from income.
 - Difficulties arise in defining “legitimate.”



Externality/Public Goods Rationales for Deviating from Haig–Simons: Charitable Giving

One reason to deviate from Haig–Simons is to **correct externalities**.

- Example: Charitable giving likely underprovided. People may give to homeless shelters because of the tax break, but these would be underfunded otherwise. Alternatively, the government could provide homeless shelters. Why not do so?
- If the government subsidizes homeless shelters, the amount of private charitable giving to those shelters would most likely fall (**crowd-out**).
- When the government tax subsidizes charitable giving, it may “**crowd in**,” or increase, private contributions.



Marginal Versus Inframarginal Effects of Tax Subsidies

Tax breaks have marginal and inframarginal effects.

Marginal impacts: Changes in behavior the government hopes to encourage through a given tax incentive.

Inframarginal impacts: Tax breaks the government gives to those whose behavior is not changed by new tax policy.



Effects of Tax Subsidies Versus Direct Spending

- Mathematically, the *government should use a tax break instead of direct spending if*: the increase in charity per \$ of tax break > 1
It implies reduction in charity per \$ of government spending.
- If this holds, then giving more of a tax break and reducing government spending increases charitable giving
- Several studies have concluded that for each 1% fall in the price of charitable giving, the amount of giving rises by 1%.



Consumer Sovereignty Versus Imperfect Information

- Direct government provision imposes preferences about how funds are spent.
- Tax subsidies to private individuals respects the preferences of citizens.
- But the private sector may not have the appropriate mechanisms in place to ensure efficient distribution of charitable spending.
 - Most individuals do little research before donating.
 - This approach has the potential to lead to gifts that don't provide much benefit to the intended target.



Taxation of households and families

- The unit of taxation matters
- Tax liabilities depend on:
 - marital status
 - number of earners
 - presence of children
- Many systems implicitly favor:
 - single-earner couples
 - households with one main earner
- Family taxation shapes work incentives within the household

Source: OECD (2023), *Taxing Wages*, OECD (2022), *Tax Policy Reforms*



The appropriate Unit of Taxation

- How should the government choose the appropriate unit of taxation?
 - Should taxes be levied on total family income or just on the incomes of individuals?
- We might like a tax system to satisfy three principles:
 - Progressivity
 - Across-Family Horizontal Equity
 - Across-Marriage Horizontal Equity
- It is impossible to achieve all three goals at once



The Problem of the “Marriage Tax” 1

- Any tax system that tries to achieve horizontal equity and progressivity has a marriage tax for some people.
 - Marriage tax: A rise in the joint tax burden on two individuals from becoming married.
- Progressivity, with taxes applied to individual income, means that two couples with different earnings distributions have different tax burdens.
- Taxing family incomes leads to a marriage tax.



The Problem of the Marriage Tax 2

Consider a tax system with:

- 10% tax rate on income up to \$20,000
- 20% marginal rate up to \$80,000
- 30% marginal rate on any income above \$80,000

Consider two couples:

	Individual Income
Yasmin Doug	140,000 10,000
Jan Elena	75,000 75,000



The Problem of the Marriage Tax 3

Total tax liability Yasmin= $10\% \times \$20,000 + 20\% \times (\$80,000 - \$20,000) + 30\% \times (\$140,000 - \$80,000) = \$32,000$

Total tax liability Doug= $10\% \times \$10,000 = \$1,000$

Total tax liability Jan = $10\% \times \$20,000 + 20\% \times (\$75,000 - \$20,000) = \$13,000$

Total tax liability Elena= $10\% \times \$20,000 + 20\% \times (\$75,000 - \$20,000) = \$13,000$



The Problem of the Marriage Tax 4

Total tax liability Family= $10\% \times \$20,000 + 20\% \times (\$80,000 - \$20,000) + 30\% \times (\$150,000 - \$80,000) = \$35,000$

Table 18-3

	Individual Income	Total Family Income	Individual Tax	Family Tax with Individual Filing	Family Tax with Total Family Income
Yasmin Doug	140,000 10,000	150,000	32,000 1,000	33,000	35,000
Jan Elena	75,000 75,000	150,000	13,000 13,000	26,000	35,000

A progressive tax system based on total family income imposes a **marriage tax** on both couples, as they both pay more in taxes as married couples than they would as singles. The ideal tax system would not distort the decision to become married



Marriage Taxes in Practice - Why We Care

- Families with equal incomes are not paying equal taxes. There is no set of deductions that would make the system of family-based taxation marriage neutral rather than providing subsidies to some marriages and taxing others.
- Society might actually want to encourage marriage, not discourage it through marriage taxes.
- A high marginal tax rate on secondary earners could reduce the labor supply of secondary earners.



Additional implication of joint taxation

- In a progressive tax system with joint taxation, the income of the secondary earner is taxed at the household's marginal tax rate
- This can reduce labor supply incentives for secondary earners
- Example:

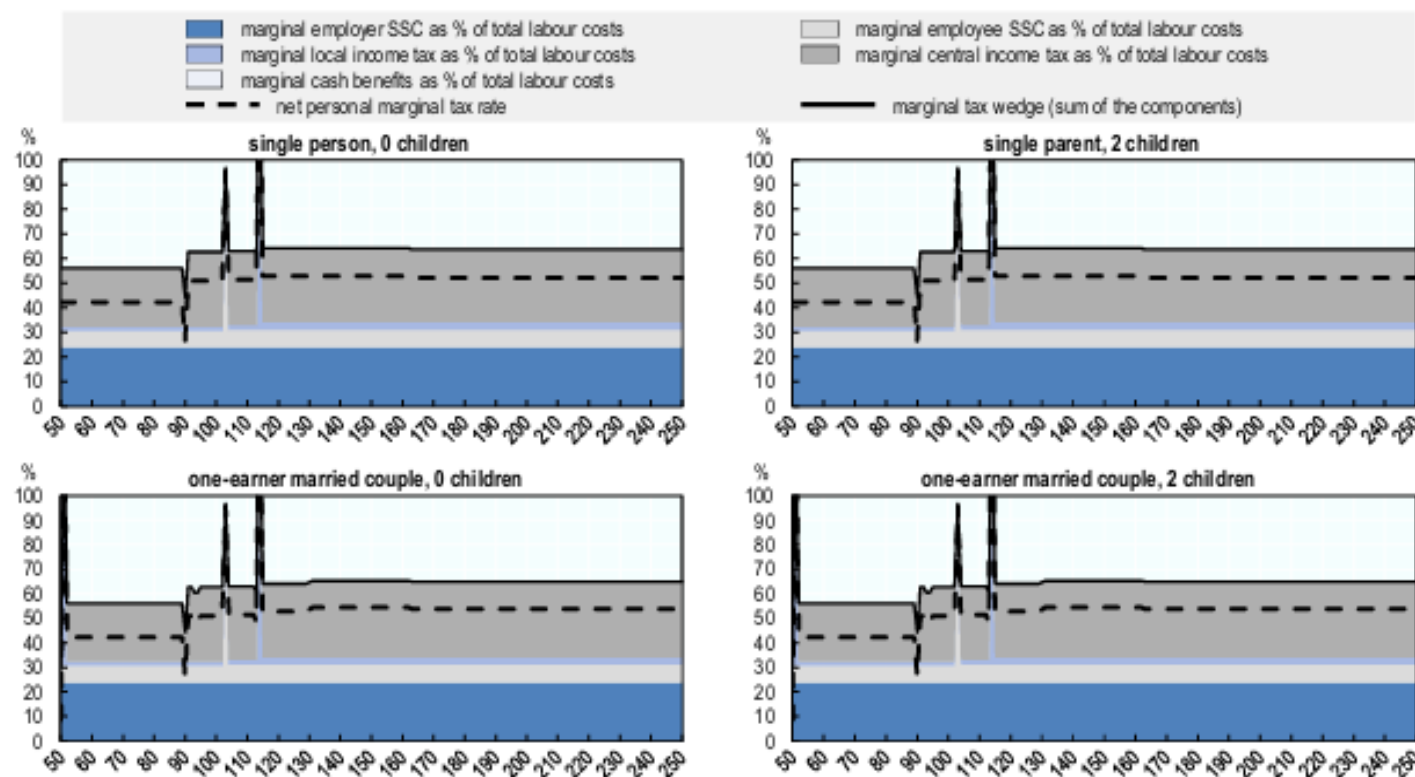
Suppose the tax rate is 20% up to €50,000 and 40% above €50,000. If one spouse earns €60,000, the household already faces the 40% marginal tax rate.

If the second spouse takes a job earning €20,000, that income may be taxed at 40%, reducing the net gain from working.



Marginal Tax Wedge Decomposition in Italy, 2022

by level of gross earnings expressed as a % of the average wage



The marginal tax wedge shows how much of an extra euro of labour cost is lost to taxes and contributions rather than received as net pay; in Italy it is high and highly non-linear, with sharp spikes driven by tax, contribution, and benefit changes that distort work incentives at specific thresholds.

Source: OECD Taxing wedges, 2023



Conclusion

- There are different types of taxes, which can be sorted into four broad categories: taxes on individual income, taxes on corporate income, taxes on wealth and taxes on consumption
- Measuring the fairness or equity of the income tax is difficult and involves considerations such as how to measure tax rates and how to define equity
- The ideal tax system would not distort the decision to become married



Exercises



MC 1

Which statement describes a progressive tax on income?

- a) Average tax rates increase with income
- b) Marginal tax rate is larger than average tax rate
- c) The share of income paid into taxes increases when the income increases
- d) All answers are correct



MC 2

According to the Haig–Simons definition, taxable income should include:

- a) Only labor income
- b) Only realized income
- c) Consumption plus saving
- d) Consumption minus saving



MC 3

Why do most real-world tax systems deviate from the Haig–Simons income base?

- a) Because it is too progressive
- b) Because it discourages labor supply
- c) Because it is difficult to measure and administer
- d) Because it raises too little revenue



MC 4

Horizontal equity requires that:

- a) High-income individuals pay higher average tax rates
- b) Individuals with different abilities pay different taxes
- c) Individuals with the same ability to pay are treated equally
- d) Tax burdens depend only on efficiency considerations



MC 5

Why can joint taxation of married couples generate a marriage tax or bonus?

- a) Because married couples always pay higher taxes
- b) Because marginal tax rates are lower for singles
- c) Because tax liabilities depend on household income only
- d) Because tax liabilities depend on how income is split within the household



True/False (please provide a justification)

- 1) Typically, taxes on income are progressive.
- 2) A tax system with a constant (flat) statutory tax rate can still be progressive if there are deductions or allowances.
- 3) Joint taxation can discourage labor supply of secondary earners.



Exercises

EXERCISE 1

Consider the following tax schedule:

- Income up to €20000: 10%
- Income between €20000 and €50000: 30%
- Income above €50000: 40%

Questions:

- Compute total tax paid by an individual earning €60000.
- Compute the average tax rate.
- Identify the marginal tax rate and explain its relevance for behavior.



Exercises

EXERCISE 2

An individual has the following economic situation in a given year:

- Labor income: €40000
- Interest income: €2000
- Increase in asset values (unrealized capital gains): €6000
- Consumption expenditure: €35000

Questions:

- Compute taxable income under a Haig–Simons income tax.
- Compute the tax base under a consumption tax.
- Explain why the two tax bases differ.

